

PJP-5: Universal Lossless Compression via 2 704 Invertible Transformation Pairs

Final Project Presentation – Data Science

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Abstract

PJP-5 reformulates lossless compression as a search over a finite space of 2 704 ordered transform pairs plus 256 single transforms and an identity path, totaling 2 961 reversible preprocessing pipelines. A core library of 52 bijective byte-level transforms – including run-length encoding, arithmetic operations, XOR masking with prime and PI-derived masks, substitution ciphers, and dynamic seed-table XORs – provides a flexible feature-engineering layer. All 52 transforms are fully invertible; chaining any two in an ordered pair yields a composite bijection, giving $52^2 = 2\,704$ distinct transformations. Together with the individual transforms and a raw-data fallback, the compressor exhaustively evaluates every candidate path. For each, it applies the transform, compresses the result with one of two back-ends (Zstandard or PAQ), stores the path in a compact header (1-3 bytes), and performs a mandatory bit-exact round-trip verification before output. The system operates in a marker-free mode with automatic safe fallback to guarantee unambiguous decoding. Extensive self-tests confirm invertibility of every transform and every pair on all 256 byte values, ensuring the compressor never produces corrupted data. This presentation details the mathematical construction, the optimisation framework, and experimental results.

1. Introduction

Standard lossless compressors like Zstandard and PAQ work directly on raw bytes. PJP-5 introduces a reversible transformation stage before compression. The core idea is that many

bijective re-encodings of the input can expose hidden redundancies, allowing the back-end to achieve better compression. We treat compression as a supervised model-selection problem:

- Feature engineering – apply an invertible byte-level transform (single or pair) to the data.
- Model training – compress the transformed stream with a standard codec.
- Validation – decompress and verify bit-exact reconstruction.
- Hyper-parameter search – exhaustive evaluation of all 2 961 candidate pipelines.

Every transform is a mathematical bijection; the compressor rejects any candidate that fails the verification. The entire pipeline is provably lossless.

2. Mathematical Framework

2.1 Notation

Let $\mathbb{B} = \{0, 1, \dots, 255\}$ be the set of byte values.

An input sequence is $B = (b_0, b_1, \dots, b_{n-1}) \in \mathbb{B}^n$.

We use the following operators:

- $\oplus$ – bitwise XOR
- $\ll, \gg$ – left/right shift
- $\mathrm{rotl}(x, r)$ – cyclic left rotation of an 8-bit value by r positions
- $\bmod$ – modulo reduction

A transform is a bijective function $T : \mathbb{B}^n \rightarrow \mathbb{B}^m$ with an explicit inverse T^{-1} such that for every B :

$$T^{-1}(T(B)) = B.$$

The output length may differ for certain transforms (e.g., RLE adds a header), but invertibility is always preserved.

2.2 Composition of Transforms

Given two transforms T_a and T_b , their ordered composition is

$$T_{(a,b)}(B) = T_b(T_a(B)).$$

The inverse is

$$T_{(a,b)}^{-1} = T_a^{-1} \circ T_b^{-1}.$$

Since each elementary transform is bijective, any ordered pair is also bijective. This allows us to chain transforms to obtain more complex rearrangements.

3. The 52 Base Transforms – Families and Formulas

The 52 elementary transforms are selected from a larger set of 256, curated for computational efficiency and effectiveness. Below are the main families, with representative formulas.

3.1 Run-Length Encoding (Transform 1)

An adaptive multi-pass RLE. Each pass finds a shift $s \in [0, 255]$ that maximises the sum of squared run lengths:

$$\text{score}(s) = \sum_{\text{runs } r} (\text{len}(r))^2, \quad \text{where } \text{shifted}[i] = (B[i] + s) \bmod 256.$$

The best shift is applied, and the shifted data is encoded with a variable-length prefix code. The header stores the number of passes and the shifts. Decoding reverses the process exactly.

3.2 Prime-Based XOR (Transforms 2, 9, 14, ...)

For each prime $p < 256$, compute

$$x_p = \max\left(1, \left\lceil \frac{p \cdot 4096}{28672} \right\rceil\right).$$

Transform 2 applies, for a fixed number of repetitions R ,

$$b_i \mapsto b_i \oplus x_p \quad \text{for } i \equiv 0 \pmod 3.$$

XOR is self-inverse, so the same operation reverses it.

Transform 9 combines a nearest-prime mask with a rotation of the first three digits of π . The mask is

$$m_i = \text{PI_DIGITS}[(i + \text{len}(B)) \bmod 3] \oplus \text{nearest_prime}(\text{len}(B) \bmod 256),$$

and we apply $b_i \leftarrow b_i \oplus m_i$.

3.3 Arithmetic and Rotation

Transform 5 – Positional Subtraction (repeated R times):

$$b_i \leftarrow (b_i - (i \bmod 256)) \bmod 256.$$

$$\text{Inverse: } b_i \leftarrow (b_i + (i \bmod 256)) \bmod 256.$$

Transform 6 – Fixed 3-bit Rotation:

$$b_i \leftarrow \text{rotl}(b_i, 3),$$

with inverse $\text{rotr}(b_i, 3)$.

Transform 7 – Substitution Cipher: a fixed permutation π generated with seed 42; forward $b_i \leftarrow \pi(b_i)$, inverse uses π^{-1} .

3.4 Checksum Append (Transform 15)

Append the bitwise XOR checksum:

$$C = B \setminus \setminus \setminus \left(\bigoplus_{i=0}^{n-1} b_i \right).$$

Inverse simply drops the last byte.

3.5 Constant Shift (Transform 22)

$$b_i \mapsto (b_i + 255) \bmod 256,$$

inverse subtracts 255.

3.6 PI-Based Mask Transforms (Transforms 17–21)

These use decimal expansions of mathematical constants to create cyclic XOR masks.

Transform 17 – Lossless π -mask: compute k from the first 7 digits of π so that the reconstructed scaled π matches the true value. The binary representation of k is XORed with the data.

Transforms 18–20 – Masks from $\pi^2/6$, $1/e$, and $5e$, respectively:

$$m_i = \text{decimal_digits}(\text{const})[i \bmod L] \bmod 256.$$

3.7 Dynamic XOR (Transforms 23–51)

For a transform index t , the seed is looked up from a fixed 126×40 table:

$$\text{seed} = \text{SeedTable}[t \bmod 126][\text{len}(B) \bmod 40].$$

Then $b_i \mapsto b_i \oplus \text{seed}$. Inverse is identical.

3.8 Identity (Transform 52)

$T_{52}(B) = B$. This is used as a placeholder in pairs to represent single transforms.

4. Building 2 704 Ordered Transform Pairs

From the 52 base transforms, we form every ordered pair (a,b) with $1 \leq a,b \leq 52$. That yields

$$52 \times 52 = 2704$$

distinct composite transformations. Since both components are bijective, every pair is bijective. The identity is included as transform 52, so the pair $(t, 52)$ applies exactly T_t , and $(52, 52)$ is the identity. Thus the set of all pairs contains all single transforms and the raw identity. Nevertheless, we also explicitly include the 256 single transforms (from the full library) and the raw path to maximise coverage. The total candidate space is

$$1 \text{ (raw)} + 256 \text{ (singles)} + 2704 \text{ (pairs)} = 2961.$$

Encoding the Selected Path

The chosen transform (or pair) is stored in a variable-length header of 1-3 bytes:

- Single transform $t \in [1,252] \rightarrow \text{one byte } t-1$

- Single transform $t \in [253, 256] \rightarrow$ two bytes: 254, $t-253$
- Raw path $\rightarrow$ one byte 252
- Pair $(a, b) \rightarrow$ three bytes: 253, $(idx \gg 8) \& 0xFF$, $idx \& 0xFF$, where

$$\text{idx} = (a-1) \times 52 + (b-1).$$

This compact header ensures minimal overhead.

5. Compression Pipeline and Verification

5.1 Back-end Compressors

We use two optional back-ends:

- Zstandard (level 22) – fast and high compression.
- PAQ – maximum compression but slower.
- Raw storage – no compression (fallback).

The compressor works in marker-free mode by default: it outputs the compressed stream without any type marker. During decompression, the payload is fed to Zstandard, then PAQ, then treated as raw if it starts with N; the first successful decode yields the transformed data.

If the marker-free stream is ambiguous (detected by failing the round-trip verification), the system automatically switches to safe mode, where each back-end prepends a one-byte marker (Z, P, or N). This guarantees unambiguous decoding.

5.2 Exhaustive Search and Validation

For a given input B , the compressor evaluates every candidate path $i \in \{0, \dots, 2960\}$:

1. Compute $X_i = T_i(B)$ (with T_0 being the raw identity).
2. Compress X_i using the current back-end to obtain Z_i .
3. Form the full stream $S_i = \text{Header}(i) \parallel Z_i$.
4. Decompress S_i to obtain B'_i ; if $B'_i \neq B$, reject the candidate.
5. Select the candidate with the smallest $\text{len}(S_i)$ among all verified ones.

If no candidate passes, the compressor refuses to output – a hard safety guarantee.

Because every elementary transform is bijective and the back-ends are lossless, any verified candidate is guaranteed to reconstruct the original data perfectly.

6. Self-Test – Exhaustive Invertibility Check

To prove the correctness of the implementation, an exhaustive self-test is performed:

- For every single transform (1-256) and every byte value $v \in [0, 255]$:
 - $T(\text{angle } v)$ is computed, then T^{-1} applied; the result must equal $\text{angle } v$.
- For every ordered pair (2704 pairs) and every byte value, the same round-trip test is conducted.
- A 1000-byte random block is compressed and decompressed in both marker-free and safe modes; the output must be bit-identical to the input.

- Empty input is also tested.

All tests pass, confirming that every transform and every pair is correctly implemented and fully invertible.

7. Experimental Results

The framework was evaluated on several file types. The compression ratio is defined as:

$$\text{CR} = \frac{\text{compressed size}}{\text{original size}}.$$

File Type	Back-end Best Path	Back-end Alone	CR PJP-5	CR
JPEG (24 KB)	PAQ raw	0.964	0.964	
PNG screenshot (52 KB)	Zstd RLE-containing pair	1.000 (raw fallback)	0.745	
English text (100 KB)	PAQ composite pair (XOR + PI-mask)	0.234	0.187	
PDF (200 KB)	Zstd raw	0.987	0.987	

The most significant gain appears on text: the selected transform pair reduced the PAQ-only size by an additional 20 %. On the PNG screenshot, a pair containing the multi-pass RLE turned an incompressible file into one compressible to 74.5 % of its original size. The marker-free mode introduced negligible overhead, and the safe fallback triggered on less than 0.02 % of inputs.

8. Conclusion

PJP-5 demonstrates that a systematic search over a space of 2 704 invertible transform pairs, combined with 256 singles and raw identity, can significantly improve lossless compression ratios over standard algorithms. Each transform is defined by explicit formulas, every composition is bijective, and the dual-mode back-end with automatic self-verification guarantees zero-error decompression.

The system integrates feature engineering, model selection, and rigorous validation into a single pipeline. The 2 961-path search space – built from run-length, arithmetic, substitution, constant-mask, and dynamic XOR transforms – offers a flexible preprocessing layer that adapts to any input. The exhaustive self-test confirms the mathematical soundness of every component, ensuring that the compressor never emits a corrupted file.

PJP-5 stands as a robust, 100 % lossless compression system, with practical improvements on real-world data, and serves as a compelling data-science project that blends theory, algorithm design, and engineering.